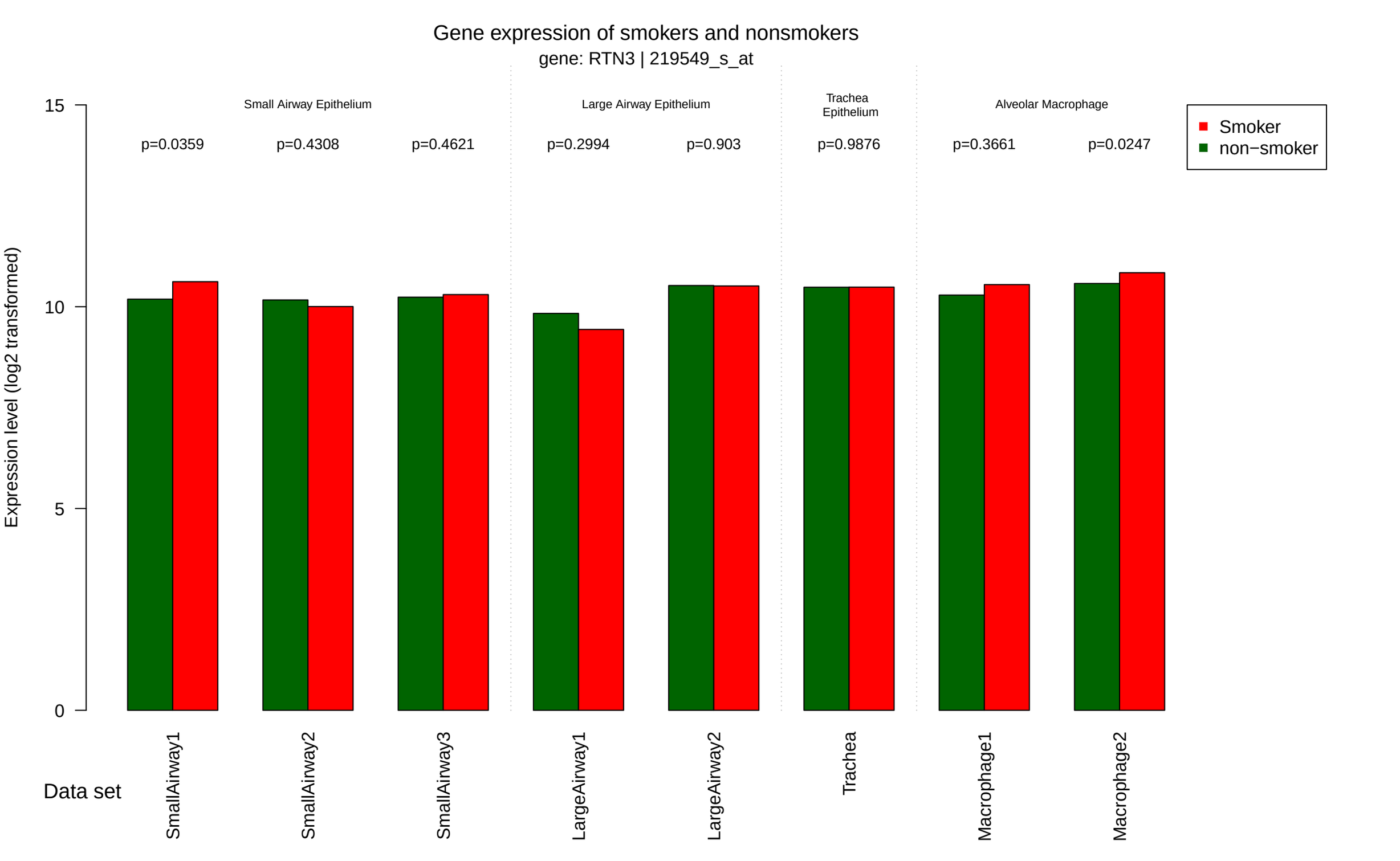
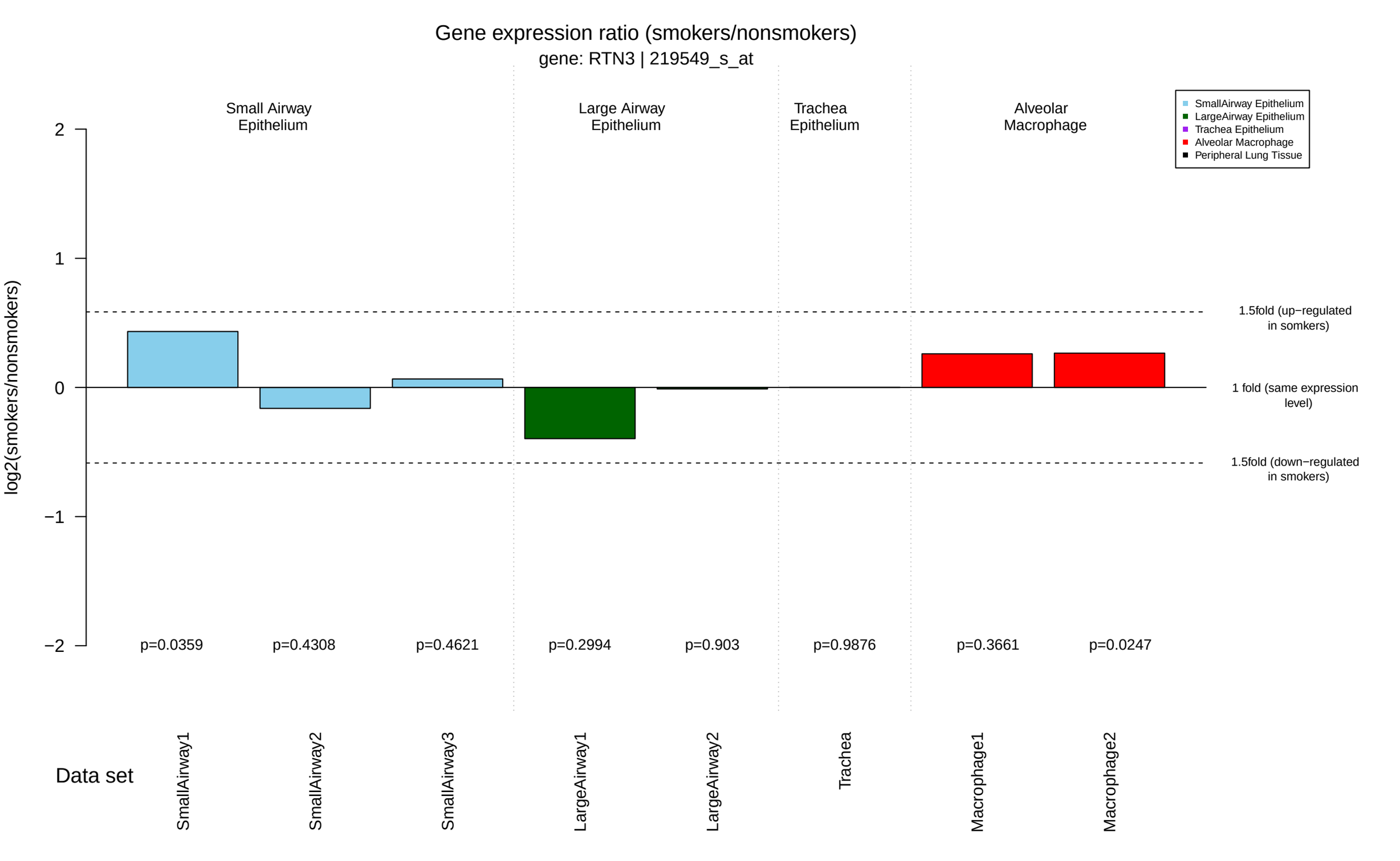
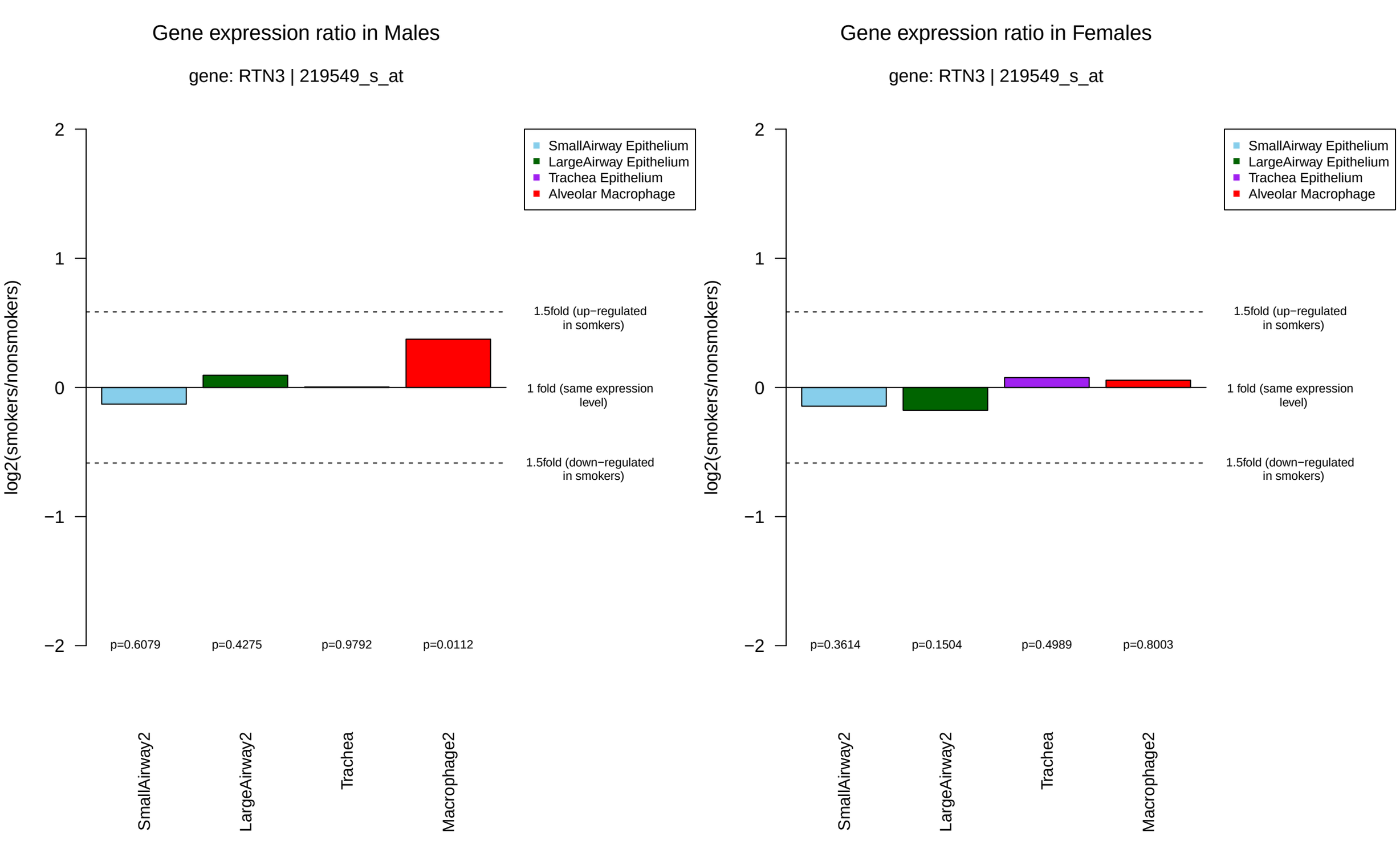
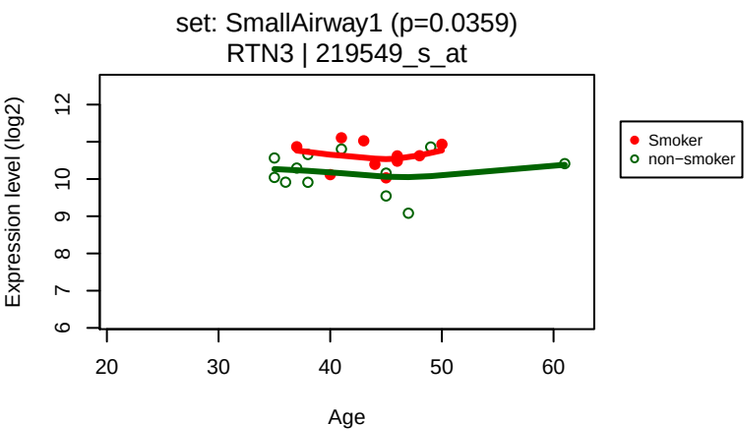
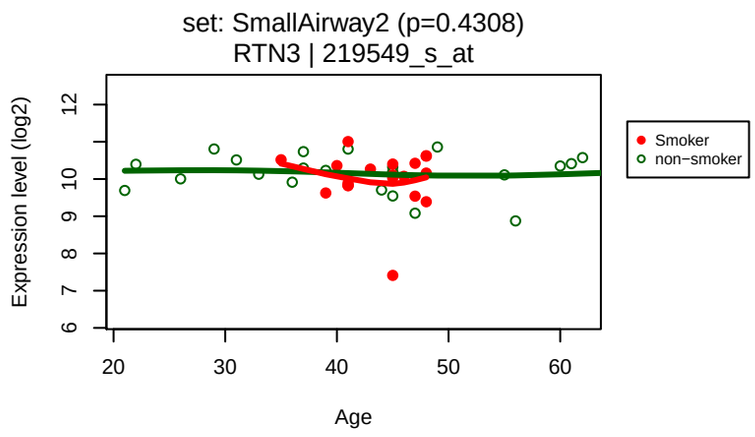
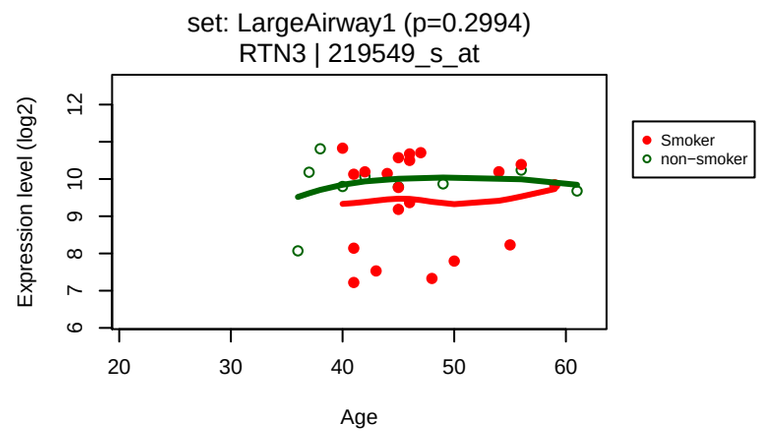

Gene expression of smokers and nonsmokers
gene: RTN3 | 219549_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0359
p=0.4308
p=0.4621
p=0.2994
p=0.903
p=0.9876
p=0.3661
p=0.0247
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

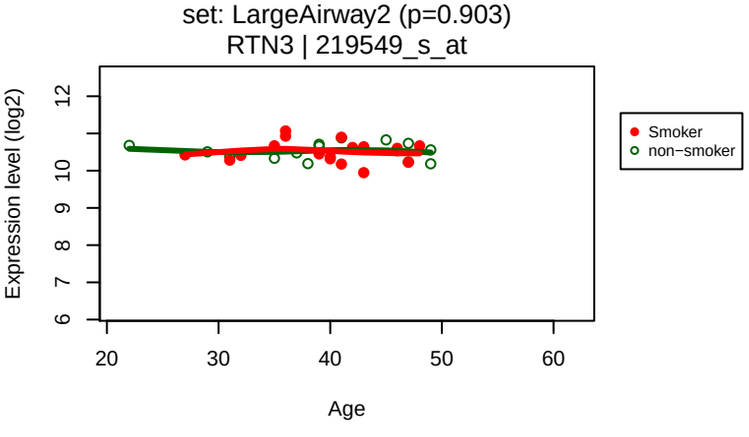

Gene expression ratio (smokers/nonsmokers)
gene: RTN3 | 219549_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0359
p=0.4308
p=0.4621
p=0.2994
p=0.903
p=0.9876
p=0.3661
p=0.0247
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

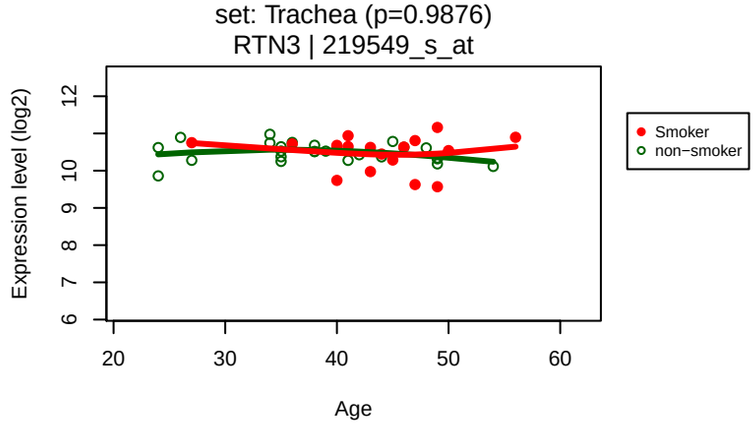

Gene expression ratio in Males
gene: RTN3 | 219549_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6079
p=0.4275
p=0.9792
p=0.0112
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN3 | 219549_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3614
p=0.1504
p=0.4989
p=0.8003
SmallAirway2
LargeAirway2
Trachea
Macrophage2

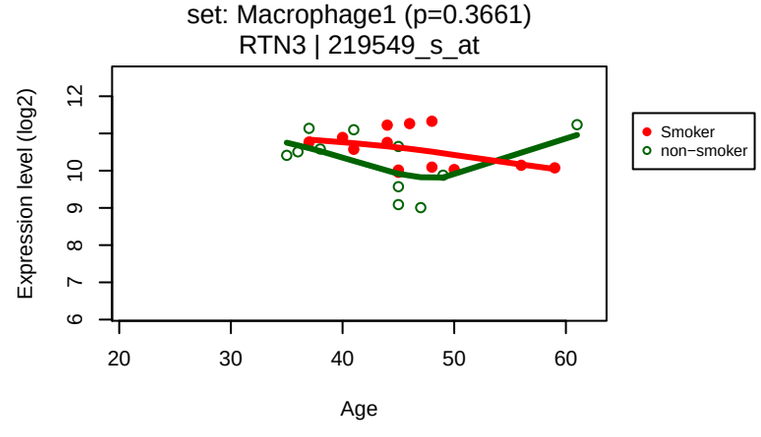

set: SmallAirway1 (p=0.0359)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker

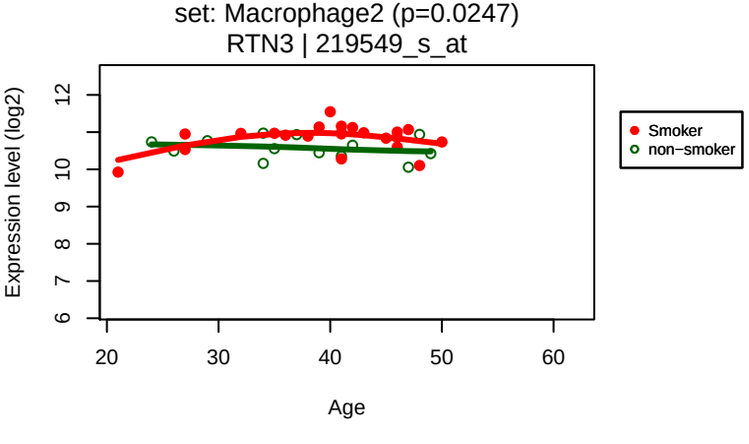

set: SmallAirway2 (p=0.4308)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker

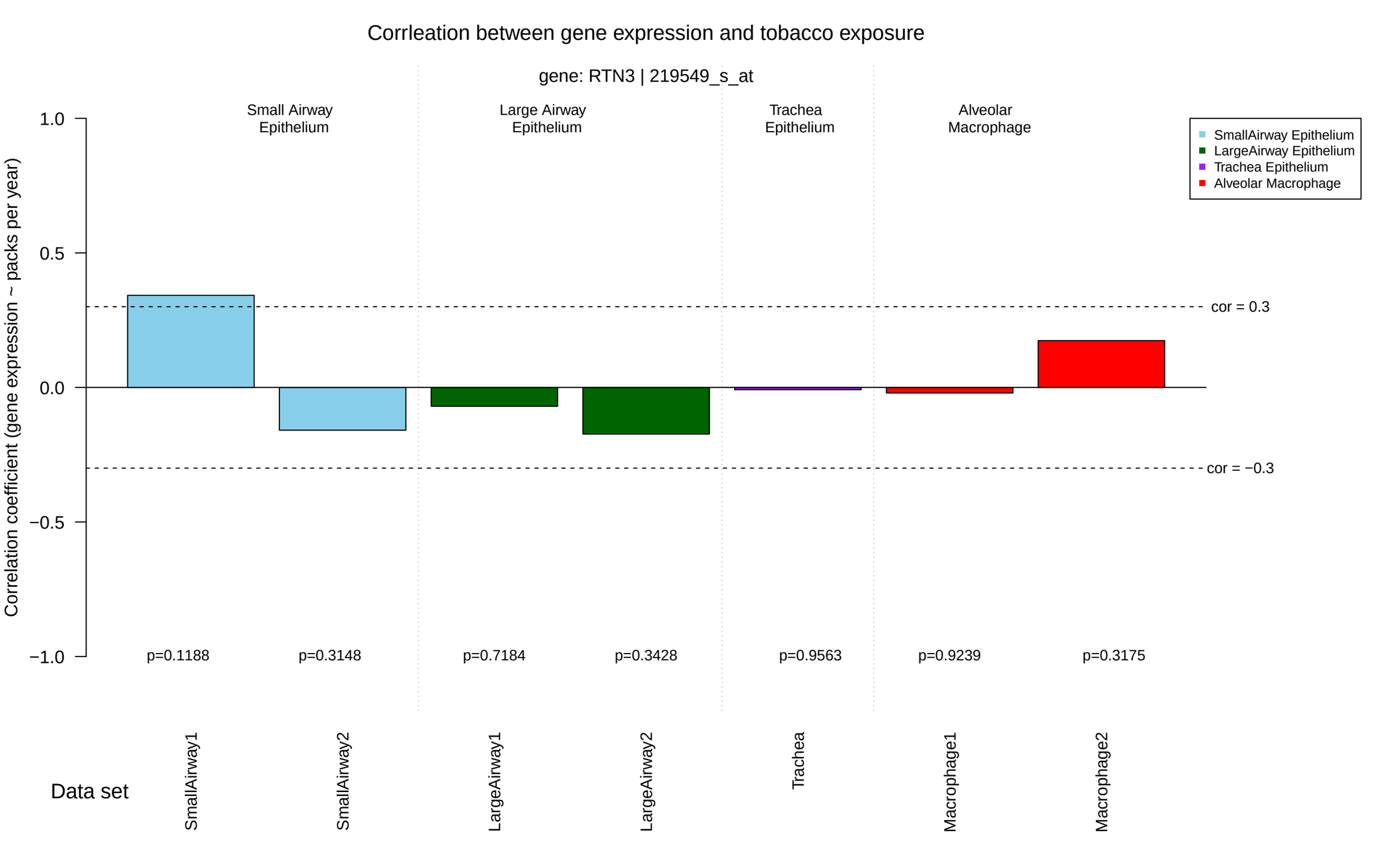

set: LargeAirway1 (p=0.2994)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.903)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9876)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3661)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0247)
RTN3 | 219549_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN3 | 219549_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1188
p=0.3148
p=0.7184
p=0.3428
p=0.9563
p=0.9239
p=0.3175
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2